ES 225 - ASSIGNMENT 3

Dataset: A suitable water-quality dataset obtained from the National Water Data Portal (NWDP), using the assigned state or region.

Q1. Statistical Analysis of Water Quality Patterns

Using the selected water-quality dataset, perform a statistical analysis of water-quality parameters across sampling locations and time, wherever the data permit.

(a) Water Quality Assessment

- Identify the chemical and physical water-quality parameters available in the dataset, such as pH, dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD), total dissolved solids (TDS), conductivity, nitrate and other available parameters.
- Calculate descriptive statistics, including the mean, median, standard deviation, minimum and maximum, for the selected parameters.
- Compare the measured values with the relevant applicable water-quality standards or criteria, where a valid standard exists for the parameter and intended water use.
- Identify parameters that may indicate potential water-quality concerns and explain their significance.

(b) Temporal or Seasonal Variation

- If sampling dates and sufficient observations are available, compare selected water-quality parameters across two seasons or time periods.
- For example, compare BOD or dissolved oxygen between two seasons, if both parameters and sufficient observations are available.
- Formulate the null (H_0) and alternative (H_a) hypotheses, perform an appropriate two-sample t-test or other justified statistical test, and report the test statistic and p-value.
- Interpret whether the observed difference is statistically significant at $\alpha = 0.05$.

(c) Spatial Variation

- Select two monitoring stations or sampling locations from the dataset.
- Compare the mean values of suitable water-quality parameters between the selected locations using an appropriate statistical test.
- Report the hypotheses, test statistics, p-values and conclusions.
- Discuss possible spatial differences in water quality, based on the available evidence.

Present the findings using tables and graphical visualizations. If the dataset lacks sufficient temporal or spatial observations, explain the limitation and perform the analyses supported by the available data.

Q2. Missing Data Analysis

- Identify missing values in the selected water-quality dataset.
- Prepare a table reporting the number and percentage of missing values in each column.
- Identify parameters with substantial missingness.

- Discuss how missing observations may affect water-quality assessment and predictive modelling.

Q3. Imputation of Missing Data

Apply the following methods to appropriate variables containing missing values:

- Mean imputation.
- Median imputation.
- Mode imputation.

Compare the original and imputed datasets by examining changes in the mean, variance and distribution of the affected parameters. Use tables and suitable visualizations to present the results.

Discuss the advantages and limitations of the three methods, particularly how imputation may affect the interpretation of water-quality measurements.

Q4. Linear Regression Analysis

Investigate the relationship between two relevant water-quality parameters available in the selected dataset.

1. Select one water-quality parameter as the response variable (Y) and another scientifically relevant parameter as the predictor variable (X).
2. Justify the choice of variables based on their environmental significance and availability in the dataset.
3. Fit a simple linear regression model: $Y = \beta_0 + \beta_1 X + \varepsilon$.
4. Calculate the regression coefficients β_0 and β_1.
5. Plot the observed data points and fitted regression line.
6. Calculate the Residual Sum of Squares (RSS).
7. Interpret the regression coefficients and discuss the environmental implications of the observed relationship.

Q5. Ridge and Lasso Regression

Select one suitable continuous water-quality parameter as the response variable (Y). Use the remaining relevant water-quality parameters as predictors (X), subject to their availability and scientific suitability.

(a) Ridge Regression

- Fit a Ridge regression model to predict the selected response variable.
- Report the regression coefficients for at least three values of the penalty parameter λ, such as 0.01, 0.1 and 1.
- Explain how the coefficients change as λ increases.

(b) Lasso Regression

- Fit a Lasso regression model using the same response variable and predictors.
- Report which predictors get eliminated (coefficients $\rightarrow 0$) as λ increases.
- Compare Ridge and Lasso regression in terms of coefficient shrinkage, interpretability and feature selection.

Standardize the predictors before fitting the models and use a consistent data-splitting and evaluation procedure.

Q6. Principal Component Analysis (PCA)

Apply PCA to the available water-quality parameters.

- Select suitable numerical parameters from the dataset and standardize them before applying PCA.
- Report the percentage of total variance explained by the first two principal components.
- Plot the observations in the PC1-PC2 space.
- Examine the component loadings and identify the parameters that contribute most strongly to each component.
- Explain what the principal components reveal about the relationships among water-quality parameters and the dominant patterns in the dataset.

Q7. Decision Tree Regression

Using the same response variable selected for Q5, train a Decision Tree regressor to predict the selected water-quality parameter.

(a) Model Development

- Select appropriate predictor variables from the available dataset.
- Train a Decision Tree regressor.
- Report the tree depth, number of leaves and three most important predictors.
- Visualize the fitted tree using a plot or a text-based representation.

(b) Model Performance

- Calculate the Mean Squared Error (MSE) on the training and test datasets.
- Interpret the results and discuss whether the model appears to overfit or generalize well.

Q8. Random Forest Regression and Model Comparison

Using the same response variable and predictor set as in Q7:

(a) Random Forest Model

- Train a Random Forest regressor using 100 trees.
- Report the feature importances.
- Plot the feature-importance ranking and compare it with the Decision Tree model.

(b) Comparative Evaluation

- Compare the training and test performance of the Decision Tree and Random Forest models using MSE and/or R^2.
- Discuss whether Random Forest improves predictive performance or reduces overfitting compared with a single Decision Tree.
- Identify the model that performs better on the test dataset and justify your conclusion.

General Instructions

1. Obtain a suitable dataset from the National Water Data Portal for the assigned state or

region.

2. Select a dataset with sufficient observations and relevant parameters for the proposed analysis. If multiple datasets are available, consider the number of observations, parameter coverage, sampling dates and data completeness when selecting the dataset.
3. Describe the dataset, including the sampling locations, period, number of observations, parameters and units, wherever available.
4. Clearly state the statistical methods, hypotheses, assumptions and interpretations wherever applicable.
5. Present the analysis using appropriate tables, plots and statistical results.
6. Use a consistent training–testing procedure for predictive models. Avoid data leakage by fitting imputation and scaling procedures using training data only.
7. Submit a structured report along with the Python code or notebook used for the analysis.

Expected outcome: A comprehensive assessment of water-quality patterns, data quality, relationships among water-quality parameters and the predictive performance of different statistical and machine-learning models.